

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS EXTENSION 2

HSC ASSESSMENT TASK 1

MARCH 2009

General Instructions

- Working time allowed – 70 minutes
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown
- Start each question on a new page
- Attempt all questions
- All questions are of equal value
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NAME: _____

Question 1	Question 2	Question 3	Total

QUESTION 1

Mark

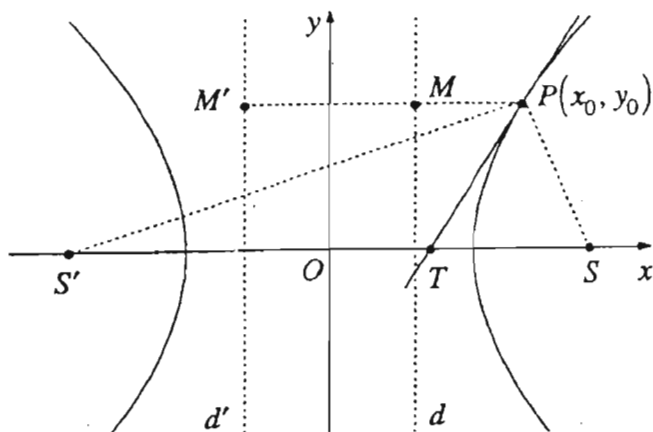
- a) Let $\alpha = 5 - 3i$ and $\beta = 2 + i$
- (i) Find $\alpha + \beta$ 1
- (ii) Find $\frac{\alpha}{\beta}$ in the form $x + iy$ 1
- (iii) If $z = x + iy$, sketch the region defined by $\text{Im}(z\alpha) < 3$ 2
- b) The complex number $z = 1 + 2i$ is a root of the equation $z^2 - aiz + b = 0$ where a and b are real numbers.
- (i) Find the values of a and b 2
- (ii) Find the other root of the equation 1
- c) Sketch the region defined by 3
- $$1 < |z - (1 + i\sqrt{3})| < 2 \text{ and } 0 \leq \arg z \leq \frac{\pi}{3}$$
- d) Let $z = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$
- (i) Express z in modulus – argument form 1
- (ii) Hence or otherwise show that z is a root of the equation $z^4 = -1$ 1
- (iii) Find the other roots of $z^4 = -1$ 2
- (iv) Find the side length of the square formed by plotting the solutions to part (iii) 1
- on an Argand diagram and joining them together.

Question 2

- a) Find the gradient of the tangent to the curve $x^3 + y^3 - 3xy = 3$ at the point $(1,2)$ 2

b)

MARK



The point $P(x_0, y_0)$ lies on the hyperbola

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

The tangent to the hyperbola at P cuts the x axis at T and has equation

$$\frac{x_0 x}{16} - \frac{y_0 y}{9} = 1$$

The two foci of the hyperbola are S and S' , and the two directrices are d and d' . The points M and M' are the closest points to P on the directrices d and d' .

- | | | |
|-------|---|---|
| (i) | Find the co ordinates of the foci | 2 |
| (ii) | Find the equations of the directrices | 1 |
| (iii) | Show that T has co ordinates $\left(\frac{16}{x_0}, 0\right)$ | 1 |
| (iv) | Using the focus- directrix definition, or otherwise, show that | 3 |

$$\frac{PS}{PS'} = \frac{TS}{TS'}$$

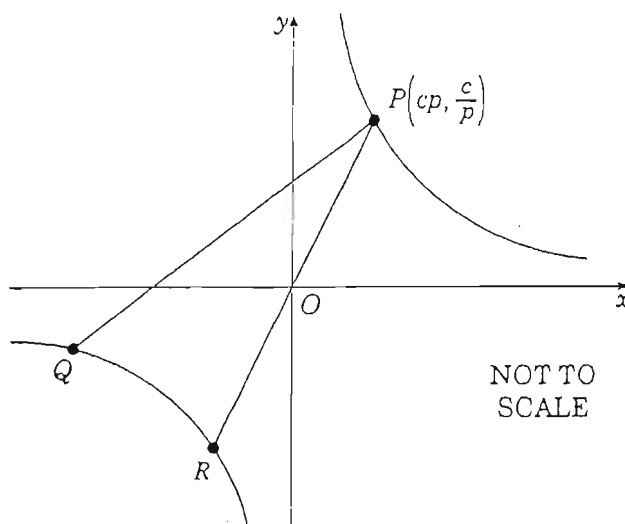
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|-----|--|---|
| (c) | Find the equation of the ellipse with eccentricity $\frac{3}{4}$ and directrices at $x = \pm 16$ | 2 |
| (d) | (i) Express $z = \frac{1+\sqrt{3}i}{1+i}$ in the form $rcis \theta$. | 3 |
| | (ii) Find the smallest positive integer n such that z^n is a real number | 1 |

QUESTION 3

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- (a) If the line $kx + my + n = 0$ is a tangent to the hyperbola $xy = c^2$, prove that $n^2 = 4c^2 km$. 2

(b)



The point $P\left(cp, \frac{c}{p}\right)$ where $p \neq \pm 1$, is a point on the hyperbola $xy = c^2$, and the normal to the hyperbola at P intersects the 2nd branch at Q . The line through P and the origin O intersects the second branch at R .

- (i) Show that the equation of the normal is
 $py - c = p^3 (x - cp)$ 2
- (ii) Show that the y coordinates of P and Q satisfy the equation.
 $py^2 - c(1 - p^4)y - p^3c^2 = 0$ 3
- (iii) Find the coordinates of Q . 1
- (iv) Show that Q, R and P are concyclic 2
- (e) (i) If w is a complex cube root of unity (ie: a root of $z^3 = 1$), prove that w^2 is also a root. 1
- (ii) Prove that $1 + w + w^2 = 0$ 1
- (iii) Hence or otherwise form a quadratic equation whose roots are given by $\alpha = 2 + w$ and $\beta = 2 + w^2$ 3

HSC Assessment Task 1 - Ext. 2

March 2009 - Solutions

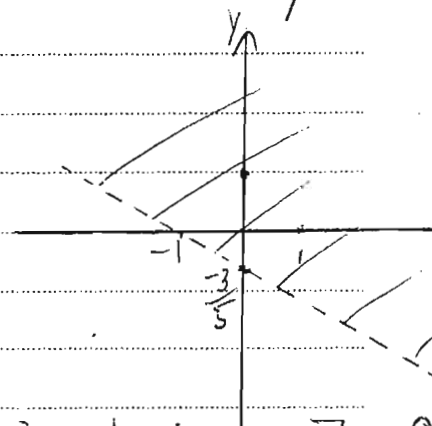
Question 1

a) $\alpha = 5 - 3i$ $\beta = 2 + i$

ci) $\alpha + \beta$
 $5 - 3i + 2 + i$
 $= 7 - 2i$

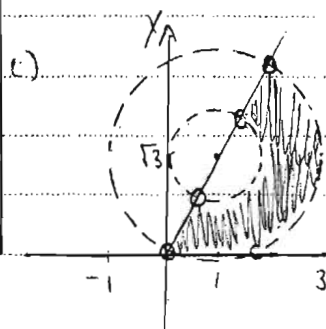
cii) $\frac{\alpha}{\beta}$
 $\frac{5 - 3i}{2 + i} \times \frac{2 - i}{2 - i}$
 $= \frac{10 - 11i - 3}{5}$
 $= \frac{7}{5} - \frac{11}{5}i$

ciii) $\text{Im}[(x + iy)(5 - 3i)] < 3$
 $\text{Im}[5x - 3xi + 5y$
 $+ 3y] < 3$
 $-3x + 5y < 3$



b) ci) $z = 1 + 2i$ is a root of
 $z^2 - aiz + b = 0$
 $\therefore (1 + 2i)^2 - ai(1 + 2i) + b = 0$
 $-1 + 4i - ai + 2a + b = 0$
 $(2a + b - 1) + i(4 - a) = 0$
 Equating real and imaginary parts to 0
 $\Rightarrow a = 4, b = -7$

cii) $z^2 - 4aiz - 7 = 0$
 Let the other root be $z = x + iy$
 Sum of roots
 $z + 1 + 2i = 4ai$
 $(x + iy) + 1 + 2i - 16i = 0$
 $(x + 1) + i(y - 14) = 0$
 $\therefore x = -1, y = 14$
 $\therefore$ Other root is $-1 + 14i$



d) ci) $|z| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2}$
 $= \sqrt{\frac{1}{2} + \frac{1}{2}}$

$|z| = 1$

$\arg z = \tan^{-1} 1 = \frac{\pi}{4}$

(ii) $Z = \text{cis } \frac{\pi}{4}$

$$Z^4 = (\text{cis } \frac{\pi}{4})^4$$

$$Z^4 = \text{cis } \pi \text{ by De Moivre's Theorem}$$

$$= -1 \text{ as req'd } \therefore \text{a root}$$

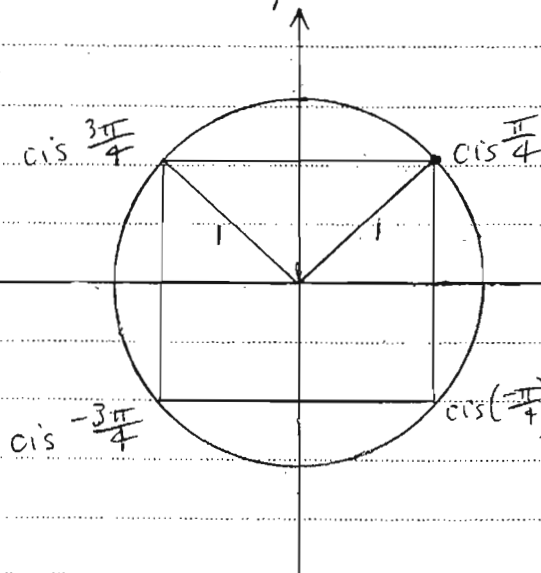
(iii) $Z^4 = -1$

$$Z^4 = \text{cis}(\pi + 2k\pi)$$

$$Z = \text{cis}\left(\frac{\pi + 2k\pi}{4}\right) \text{ by De Moivre's}$$

$$\therefore Z = \text{cis } \frac{\pi}{4}, \text{cis } \frac{-\pi}{4}, \text{cis } \frac{3\pi}{4}, \text{cis } \frac{-3\pi}{4}$$

(iv)



By Pythagoras
 $\sqrt{1^2 + 1^2}$

$\sqrt{2}$ is side length

Question 2

a) $x^3 + y^3 - 3xy = 3$

Differentiating implicitly

$$3x^2 + 3y^2 \frac{dy}{dx} - 3x \frac{dy}{dx} - 3y = 0$$

$$\frac{dy}{dx}(3y^2 - 3x) = 3y - 3x^2$$

$$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

$$\therefore \text{At } (1, 2),$$

$$m_{\text{tangent}} = \frac{1}{3}$$

b) (i) Foci at $(\pm ae, 0)$

$$a = 4$$

$$b^2 = a^2(e^2 - 1)$$

$$9 = 16(e^2 - 1)$$

$$\frac{9}{16} = e^2 - 1$$

$$e^2 = \frac{25}{16}$$

$$e = \frac{5}{4}$$

$$\therefore (\pm ae, 0) \Rightarrow (\pm 5, 0)$$

(ii) Directrices $x = \pm \frac{a}{e}$

$$x = \pm \frac{4}{5/4}$$

$$x = \pm \frac{16}{5}$$

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ciii) T on tangent is
where $y = 0$

$$\text{ie: } \frac{x_0 - \frac{16}{x_0}}{16} - 0 = 1$$

$$x = \frac{16}{x_0}$$

$$\therefore T \text{ is } \left(\frac{16}{x_0}, 0 \right)$$

civ) Need to show

$$\frac{PS}{PS'} = \frac{TS}{TS'}$$

Since $PS = e PM$ and
 $PS' = e PM'$

$$\therefore \frac{PM}{PM'} = \frac{TS}{TS'}$$

$$\frac{x_0 - \frac{16}{x_0}}{x_0 + \frac{16}{x_0}} = \frac{5 - \frac{16}{x_0}}{\frac{16}{x_0} + 5}$$

$$\frac{5x_0 - 16}{5x_0 + 16} = \frac{5x_0 - 16}{5x_0 + 16} \quad \checkmark$$

Result shown

d) $16 = \frac{a}{e}$

$$e = \frac{3}{4}$$

$$\therefore a = 12$$

$$b^2 = a^2(1 - e^2)$$

$$b^2 = 144(1 - (\frac{3}{4})^2)$$

$$b^2 = 144 \times \frac{7}{16}$$

$$b^2 = 63$$

$\therefore$ Ellipse has eq'n

$$\frac{x^2}{144} + \frac{y^2}{63} = 1$$

dci) $z = \frac{1 + \sqrt{3}i}{1 + i}$

$$1 + \sqrt{3}i = 2 \operatorname{cis} \frac{\pi}{3}$$

$$1 + i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$$

$$\therefore z = \sqrt{2} \operatorname{cis} \frac{\pi}{12}$$

cii) $z^n = (\sqrt{2})^n \operatorname{cis} \left(\frac{\pi}{12} \right)$

$$= (\sqrt{2})^n \operatorname{cis} \left(\frac{\pi n}{12} \right)$$

is real if

$$\sin \left(\frac{\pi n}{12} \right) = 0$$

ie: when $n = 12, n > 0$

Question 3

a) Solve simultaneously

$$kx + my + n = 0$$

$$xy = c^2$$

$$y = \frac{c^2}{x}$$

$$\Rightarrow kx + m \times \frac{c^2}{x} + n = 0$$

$$kx^2 + nx + mc^2 = 0$$

$\rightarrow$ If line is a tangent, only one solution $\Rightarrow \Delta = 0$

$$n^2 - 4 \times k \times mc^2 = 0$$

$$n^2 = 4kmc^2 \text{ as required}$$

b) (i) $x = \frac{c}{p}$ $\frac{dx}{dp} = -\frac{c}{p^2}$

$y = \frac{c}{p}$ $\frac{dy}{dp} = -\frac{c}{p^2}$

$\therefore \frac{dy}{dx} = \frac{\frac{dy}{dp}}{\frac{dx}{dp}} = \frac{-\frac{c}{p^2}}{-\frac{c}{p^2}} = 1$

$\frac{dy}{dx} = \frac{-1}{p^2}$

$y - y_1 = m(x - x_1)$

$y - \frac{c}{p} = \frac{p^2}{1}(x - \frac{c}{p})$

is the eq'n of the normal

$py - c = p^3(x - \frac{c}{p})$

as required

(ii) Solving the hyperbola simultaneously with the normal:

$$\begin{cases} xy = c^2 \\ py - c = p^3(x - \frac{c}{p}) \end{cases}$$

$$py - c = p^3(\frac{c^2}{y} - \frac{c}{p})$$

$$py^2 - cy = p^3(c^2 - cpy)$$

$$py^2 - cy = p^3c^2 - c p^4 y$$

$$py^2 - c(1 - p^4)y - p^3c^2 = 0$$

(iii) One root of (ii) is $\frac{c}{p}$. Let y value of be α . Product of roots is $-\frac{p^3c^2}{p}$

$$\alpha \times \frac{c}{p} = -\frac{p^3c^2}{p}$$

$$\therefore \alpha = -cp^3$$

$$\therefore x \text{ value of } Q \text{ is } \frac{c^2}{\alpha}$$

$$= \frac{c^2}{-cp^3} = -\frac{c}{p^3}$$

$$\therefore Q \left(-\frac{c}{p^3}, -c \right)$$

(iii) cont'd.

QRP are concyclic if $\angle QRP = 90^\circ$ (Angle in a semi-circle is 90° , ie: $M_{QR} \times M_{RP} = -1$)

Since $xy = c^2$ is odd R is $(-cp, -\frac{c}{p})$

$$M_{QR} = \frac{-\frac{c}{p} + \frac{c}{p}}{-\frac{c}{p^3} + cp} = \frac{\frac{c}{p} - \frac{c}{p^3}}{p - \frac{1}{p^3}}$$

$$M_{RP} = \frac{\frac{c}{p} + \frac{c}{p}}{cp + cp} = \frac{\frac{2c}{p}}{2cp} = \frac{1}{p^2}$$

ciii) cont'd

$$M_{QR} \times M_{RP}$$

$$= \frac{\frac{1}{p} - p^3}{p - \frac{1}{p^3}} \times \frac{1}{p^2}$$

$$= \frac{\frac{1}{p} - p^3}{p^3 - \frac{1}{p}}$$

$$= -1 \quad \therefore QRP \text{ are concyclic}$$

e. ci) If ω is a root of $z^3 = 1$, then $\omega^3 = 1$. If ω^2 is also a root then

$$\begin{aligned} (\omega^2)^3 &= 1 \\ \Rightarrow (\omega^3)^2 &= 1 \\ 1^2 &= 1 \quad \checkmark \quad \therefore \omega^2 \text{ is a root.} \end{aligned}$$

cii) The 3 roots of $z^3 = 1$ are 1, ω and ω^2 .

$$\begin{aligned} \text{Sum of roots} &= 1 + \omega + \omega^2 \\ &= -\frac{b}{a} \text{ from polynomial theory} \end{aligned}$$

$$\begin{aligned} z^3 - 1 &= 0 \\ -\frac{b}{a} &= 0 \end{aligned}$$

$$\therefore 1 + \omega + \omega^2 = 0 \text{ as required.}$$

ciii) Roots of quadratic are α and β 3

$$z^2 - (\alpha + \beta)z + \alpha\beta = 0$$

$$z^2 - (2 + \omega + 2 + \omega^2)z + (2 + \omega)(2 + \omega^2) = 0$$

$$z^2 - (1 + \omega + \omega^2 + 3)z + (4 + 2\omega^2 + 2\omega + \omega^3) = 0$$

$$z^2 - 3z + [5 + 2(\omega + \omega^2)] = 0 \quad \text{since } \omega^3 = 1 \text{ and using cii)}$$

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$$z^2 - 3z + (5 + 2(-1)) = 0$$

$$z^2 - 3z + 3 = 0$$